Problem Set 4

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Question 1:

```r
# Load necessary libraries
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.1.4     v readr     2.1.5
v forcats   1.0.0     v stringr   1.5.1
v ggplot2   3.5.2     v tibble    3.3.0
v lubridate 1.9.4     v tidyr     1.3.1
v purrr     1.1.0
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becon
```

```r
library(haven)
library(here)
```

```
here() starts at /Users/tate/SchoolWork
```

```r
library(stargazer)
```

```
Please cite as:

 Hlavac, Marek (2022). stargazer: Well-Formatted Regression and Summary Statistics Tables.
 R package version 5.2.3. https://CRAN.R-project.org/package=stargazer
```

```r
library(broom)
library(modelsummary)
library(psych)
```

```

Attaching package: 'psych'

The following object is masked from 'package:modelsummary':

    SD

The following objects are masked from 'package:ggplot2':

    %+%, alpha
```

```r
# Load the dataset
mc_dat <- read_dta(here("Y2S1/Metrics/PSets/PS4/data", "medicarePS.dta"))

describe(mc_dat)
```

```
           vars     n  mean   sd median trimmed  mad   min   max range  skew
DelayCare     1 63165  0.07 0.25   0.00    0.00 0.00  0.00  1.00   1.0  3.42
Ninsurance    2 63165  1.25 0.59   1.00    1.28 0.00  0.00  2.00   2.0 -0.12
Age           3 63165 63.88 5.73  63.25   63.68 7.41 55.25 74.75  19.5  0.23
Minority      4 63165  0.28 0.45   0.00    0.22 0.00  0.00  1.00   1.0  1.00
Education     5 63165  2.15 0.81   2.00    2.19 1.48  1.00  3.00   2.0 -0.28
           kurtosis   se
DelayCare      9.69 0.00
Ninsurance    -0.49 0.00
Age           -1.16 0.02
Minority      -1.00 0.00
Education     -1.42 0.00
```

```r
summarise(mc_dat)
```

```
# A tibble: 1 x 0
```

Part A

Becuase at 65 all individuals are eligible for Medicare, it inherently acts as a treatment on `Ninsurance` (the number of insurance plans possessed by an agent). So, using 65 as the threshold for RDD is appropriate due to: 1. The amount `Ninsurance` changes discontinuously at age 65, seeing agents go from at least zero to 1, or from 1 to 2. 2. It is fair to assume that individuals just before and just after their 65th birthday are similar in all other respects, meaning that any discontinuous change in the outcome variable can be ascribed to the treatment. 3. The running variable (age) is continuous around the threshold of 65.

Part B

```
# Plotting averages per quarter of age: outcome, first stage, validity
mc_dat <- mc_dat %>%
  mutate(age_quarter = floor(Age * 4) / 4)
```